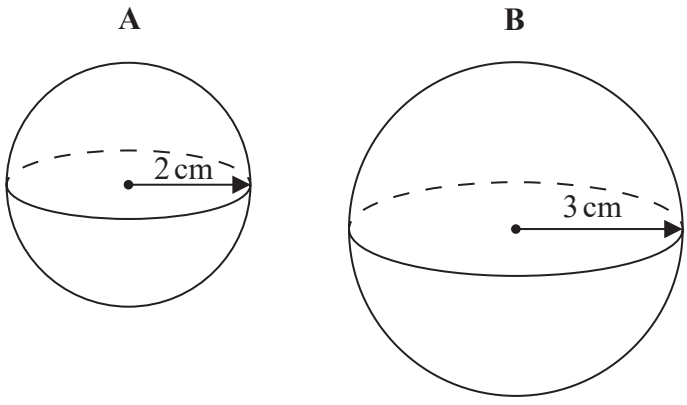
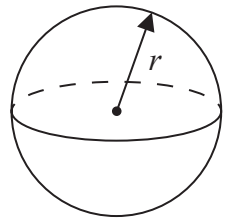
16 Here are two solid spheres, **A** and **B**.



Volume of sphere $= \frac{4}{3}\pi r^3$



Sphere **A** is made of gold.
Sphere **B** is made of silver.

Sphere **A** has radius 2 cm.
Sphere **B** has radius 3 cm.

Gold has a density of 19 000 kg/m^3
Silver has a density of 10 000 kg/m^3

Which sphere has the greater mass?
You must show how you get your answer.

Density:19000kg/m^3 = 19000×1000÷1000000

=19g/cm^3; silver similarly10g/cm^3.

1 mark
Convert density units

Sphere volumes: A = (4π/3)×2^3; B = (4π/3)×3^3.

1 mark
Sphere volume

Mass A = (4π/3) × 8 × 19 = (4π/3) × 152g.

1 mark
Volume × density

Mass B = (4π/3) × 27 × 10 = (4π/3) × 270g.

Sphere B,silver:270 >152 and4π/3 is positive.

1 mark
Compare both masses

(Total for Question 16 is 4 marks)